

Prove the relation :

X has sparsity level $s \in \{2, \dots, d\}$

$$\delta_{PW}(X) \leq \delta_s(X) \leq s \delta_{PW}(X)$$

Proof:

$$\delta_{PW}(X) = \max_{i,j \in \{1, \dots, d\}} \left| \frac{\langle X_i, X_j \rangle}{n} - \mathbb{1}[i=j] \right|$$

$$\delta_s(X) = \max_{\substack{S \subseteq \{1, \dots, d\} \\ |S| \leq s}} \left\| \frac{X_S^T X_S}{n} - I_{|S|} \right\|_2$$

X_S : concatenation of columns of X with indices inside S .

i) lower bound.

$\forall (i, j), S = \{i, j\} :$

$$A_S = \frac{X_S^T X_S}{n} - I_2 =$$

$$\begin{bmatrix} \frac{\langle X_i, X_i \rangle}{n} - 1 & \frac{\langle X_i, X_j \rangle}{n} \\ \frac{\langle X_j, X_i \rangle}{n} & \frac{\langle X_j, X_j \rangle}{n} - 1 \end{bmatrix}$$

entries of A_S are of the form $\frac{\langle X_i, X_j \rangle}{n} - \mathbb{1}[i=j]$

entries of A_S are bounded from above by $\delta_{PW}(X)$

* Spectral norm of a matrix M is at least the magnitude of any entry.

$$\Rightarrow \left\| \frac{X_s^T X_s}{n} - I_2 \right\|_2 \geq \max_{i,j} \left| \frac{\langle X_i, X_j \rangle}{n} - \mathbb{1}[i=j] \right| = \delta_{pw}(X)$$

$$\delta_2(X) \geq \delta_{pw}(X)$$

$$\delta_s \text{ is non-decreasing in } s. \Rightarrow \delta_s(X) \geq \delta_{pw}(X) \quad \checkmark$$

ii) upper bound.

Gershgorin's disc theorem. (row version)

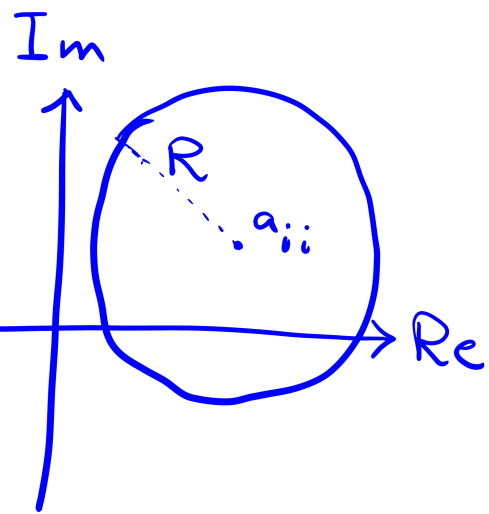
Let $A \in \mathbb{C}_{n \times n}$. For each row i , define the Gershgorin disc D_i in the complex plane as:

$$D_i = \left\{ z \in \mathbb{C} \mid |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \right\}$$

$$A = \begin{bmatrix} a_{11} & \dots & a_{1i} & \dots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \dots & a_{ii} & \dots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & a_{ni} & \dots & a_{nn} \end{bmatrix}$$

$$R = \sum_{j \neq i} |a_{ij}|$$

then every eigenvalue λ of A lies
in at least one of the discs. i.e. $\lambda \in \bigcup_{i=1}^n D_i$



Intuition:

Consider λ s.t. $Ax = \lambda x$. ($x \neq 0$) (*)

$x = (x_1, \dots, x_n)$ $k = \arg \max_j |x_j|$ $|x_k| \geq |x_j|$ for each j

$$\sum_{j=1}^n a_{kj} x_j = \lambda x_k \iff \sum_{j \neq k}^n a_{kj} x_j = (\lambda - a_{kk}) x_k$$

$$\Rightarrow |\lambda - a_{kk}| |x_k| = \left| \sum_{j \neq k}^n a_{kj} x_j \right| \overset{\text{triangle}}{\leq} \sum_{j \neq k}^n |a_{kj}| |x_j|$$

$$\overset{(*)}{\leq} \sum_{j \neq k}^n |a_{kj}| |x_k|$$

omit $|x_k|$
 $\Rightarrow |\lambda - a_{kk}| \leq \sum_{j \neq k}^n |a_{kj}| \Rightarrow \lambda \in D_k$

Back to our proof. (Prove $\delta_S(X) \leq \delta_{PW}(X)$)

$$|S| = m$$

$$A_S := \frac{X_S^T X_S}{n} - I_m$$

$$\Rightarrow |(A_S)_{i,j}| = \left| \frac{\langle X_i, X_j \rangle}{n} - \mathbb{1}_{[i=j]} \right| \overset{(\star\star\star)}{\leq} \delta_{PW}(X)$$

$\downarrow$
 $i, j \in S$

Using the theorem: λ an eigenvalue of A_S

$$|\lambda - (A_S)_{kk}| \stackrel{(**)}{\leq} \sum_{j \neq k} |(A_S)_{kj}| \quad \text{for some index } k.$$

$$A_S \text{ is symmetric} \Rightarrow \|A_S\|_2 = \max_{\lambda} |\lambda| \Rightarrow \delta_S(X) = \max_{\lambda} |\lambda|$$

$\downarrow$
eigen values of A_S

$\Rightarrow$ encourages triangle inequality.

$$|\lambda| \leq |\lambda - (A_S)_{kk}| + |(A_S)_{kk}|$$

$$\stackrel{(**)}{\leq} \sum_{j \neq k} |(A_S)_{kj}| + |(A_S)_{kk}| = \sum_{j=1}^{|S|} |(A_S)_{kj}|$$

$$\leq |S| \delta_{pw}(X)$$

$$\left\| \frac{X_S^T X_S}{n} - I_{|S|} \right\| = \|A_S\|_2 = \max_{\lambda} |\lambda| \leq |S| \delta_{PW}(X)$$

$$\Rightarrow \delta_s(X) = \max_{|S| \leq s} \left\| \frac{X_S^T X_S}{n} - I_{|S|} \right\| \leq s \delta_{PW}(X)$$

Examples for tightness. (Goal: Designing a matrix X)

i) lower bound.

$$G = \frac{X^T X}{n} = I_d + \rho \begin{bmatrix} 0 & 1 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} \quad \rho \in (0, 1)$$

X exists because $G \succeq 0$ and symmetric.

Using Cholesky $\Rightarrow \exists \tilde{X}$ s.t. $G = \tilde{X}^T \tilde{X} \Rightarrow X = \sqrt{n} \tilde{X}$

eigenvalues of $G - I_d$ are $\rho, -\rho$ and $n-2$ zeros.

$$\Rightarrow \delta_s(X) = \max_{\lambda} |\lambda| = |\rho|$$

$$\delta_{PW}(X) = \max \left| \frac{\langle x_i, x_j \rangle}{n} - \mathbb{1}[i=j] \right| = |\rho|$$

$$\Rightarrow \delta_s(X) = \delta_{PW}(X)$$

ii) upper bound.

$$G = \frac{X^T X}{n} = I_S + \rho \mathbb{1} \mathbb{1}^T$$

all ones vector. $\in \mathbb{R}^S$

for X to exist G should be PSD. $\Rightarrow$ pick ρ small enough.
(****)

$$\delta_s(X) = \|G - I_s\|_2 = |\rho| \|\underline{1} \underline{1}^T\|_2 = |\rho| \max_{\lambda_1} |\lambda_1|$$

where λ_1 eigenvalues of $\underline{1} \underline{1}^T$.

$$(\underline{1} \underline{1}^T) \underset{\downarrow x}{\underline{1}} = \underline{1} (\underline{1}^T \underset{\downarrow x}{\underline{1}}) = \underline{1} s = s \underline{1} \Rightarrow s \text{ is an eigenvalue.}$$

suppose $x \perp \underline{1}$. then:

$$(\underline{1} \underline{1}^T) x = \underline{1} \underline{1}^T x = 0 \Rightarrow 0 \text{ is an eigenvalue of } \underline{1} \underline{1}^T$$

$\underline{1}$ defines a subspace of $\mathbb{R}^s$ with $\dim = 1$.
 $\Rightarrow$ any vector orthogonal to $\underline{1}$ creates an eigenvector.

$\Rightarrow$ subspace created by eigenvectors of 0 has $\dim = s-1$

$\Rightarrow$ occurrence of 0 is $s-1$.

$$\Rightarrow \delta_s(X) \overset{(\text{****})}{=} |P|s = s \delta_{pw}(X) \quad \checkmark$$